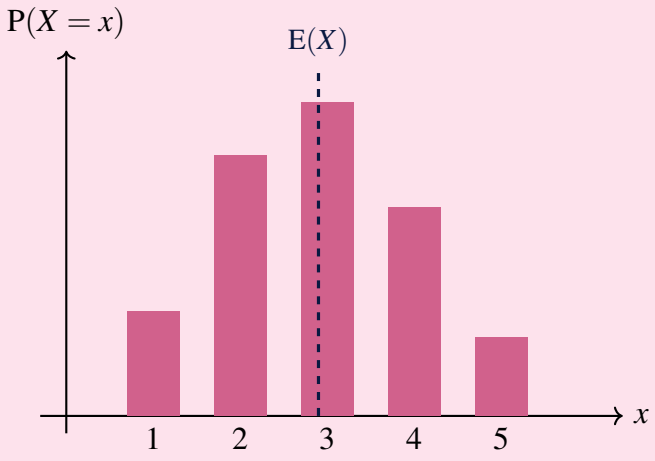
Discrete Probability Distributions

IB Maths AA SL 4.7 · Notes · Questions · Worked Answers

Probability tables · Finding unknowns · Expected value $\mathrm{E}(X)$
Fair games · Distributions from dice and counters



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= no calculator = calculator allowed

Sources: **Worksheet** = AA SL *Discrete Probability Distributions* worksheet (Revision Village questions); **RV** = Revision Village (Practice Set 2 numbering); **Oxford** = Oxford IB textbook Ex 14B and exam-style questions; **School HW** = school homework extension problems; **Practice** = written for this booklet. Repeated questions appear only once.

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1. Finding Unknowns and E(X)

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Key Notes

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1. Discrete random variables

A **discrete random variable** X takes separate values (e.g. 0, 1, 2, ...). Its **probability distribution** lists every value x with its probability:

x	0	1	2	3
$\mathrm{P}(X=x)$	0.1	0.3	0.4	0.2

Two rules: (1) $0 \le \mathrm{P}(X=x) \le 1$ (2) $\sum \mathrm{P}(X=x) = 1$

$\mathrm{P}(X>2)$, $\mathrm{P}(X \le 1)$, ...: add the probabilities in the table for those values.

2. Expected value (the mean)

$$\mathrm{E}(X) = \sum x\mathrm{P}(X=x) \qquad \text{(in the IB formula booklet)}$$

Multiply each value by its probability and add. For the table above:
$\mathrm{E}(X) = 0(0.1) + 1(0.3) + 2(0.4) + 3(0.2) = 1.7$.

- $\mathrm{E}(X)$ is a long-run average — it need not be one of the values of X.
- Expected number of times in n trials $= n \times p$; expected total over n games $= n \times \mathrm{E}(X)$.

3. Finding unknown probabilities

- **One unknown** (q, k, ...): use $\sum \mathrm{P} = 1$.
- **Two unknowns** (p and q): use $\sum \mathrm{P} = 1$ **and** the given $\mathrm{E}(X)$ (or a given probability) — solve simultaneously.
- **Formula** $\mathrm{P}(X=x) = kx$ or $k(x^2+1)$: substitute each x, add, set equal to 1.
- **Quadratic** (e.g. p and $5p^2$ in the table): solve, then **reject** any value that is negative or makes a probability greater than 1.

4. Fair games

A game is **fair** when the expected gain of the player is zero:

$$\text{E(gain)} = 0 \quad \Longleftrightarrow \quad \text{E(amount received)} = \text{cost to play.}$$

- Decide whether the table shows *points/winnings* (set E$=0$) or *money received* (set E $=$ entry fee).
- A negative E(gain) means the player is expected to lose (good for the organiser).

Example: pay \$20; win \$$k$ with probability 0.3, nothing otherwise. Fair: $0.3k = 20 \Rightarrow k = 66.67$.

5. Building a distribution from outcomes

For two dice, draw a **sample-space grid** (36 equally likely outcomes) and fill each cell with the value of X (sum, difference, minimum, ...). Then count:

$$\text{P}(X = x) = \frac{\text{number of cells showing } x}{36}$$

Check that the probabilities add up to 1.
Without replacement: use a tree diagram (the second probability changes).

Minimum of two dice

	1	2	3	4	5	6
1	1	1	1	1	1	1
2	1	2	2	2	2	2
3	1	2	3	3	3	3
4	1	2	3	4	4	4
5	1	2	3	4	5	5
6	1	2	3	4	5	6

$\text{P}(X = 2) = \frac{9}{36}$

6. TI-Nspire CX: E(X) in one go

1. home → **Add Lists & Spreadsheet**; values in column A, probabilities in column B.
2. menu → **4: Statistics** → **1: Stat Calculations** → **1: One-Variable Statistics**.
3. X1 List: a[ ], Frequency List: b[ ]. Then $\bar{x} = \text{E}(X)$ (also check Σx of b = 1).

Solve for an unknown: menu → **3: Algebra** → **1: Solve**, e.g. solve$(5p^2 + p - 0.4 = 0, p)$.

1 answer →

Calculator **Worksheet · Q1**

[Maximum mark: 6]

A random variable X has a probability distribution given in the following table.

x	0	1	2	3	4
$\mathrm{P}(X = x)$	$\frac{3}{10}$	q	$\frac{1}{10}$	$\frac{3}{20}$	$\frac{1}{4}$

(a) Find the value of q. [3]

(b) Find $\mathrm{E}(X)$. [3]

2 answer →

No calculator **Worksheet · Q2**

[Maximum mark: 6]

A random variable X has a probability distribution given in the following table.

x	0	1	2	3	4	5
$\mathrm{P}(X = x)$	p	q	0.1	0.2	0.1	r

(a) Given that $\mathrm{P}(X > 2) = 0.6$, find the value of r. [2]

(b) Given that $\mathrm{E}(X) = 2.9$, find the value of p and the value of q. [4]

1 ← question *Worksheet · Q1*

(a) $\frac{3}{10}+q+\frac{1}{10}+\frac{3}{20}+\frac{1}{4}=1$ **M1**

$q=1-0.8$ **A1**

$\boxed{q=\frac{1}{5}=0.2}$ **A1**

(b) $\mathrm{E}(X)=0(0.3)+1(0.2)+2(0.1)+3(0.15)+4(0.25)$ **M1 A1**

$=0.2+0.2+0.45+1=\boxed{1.85}$ $\left(\frac{37}{20}\right)$ **A1**

TI-Nspire CX: Question 1

Expected value: home → **Add Lists & Spreadsheet**. Type the x-values in column A and the probabilities in column B. Then menu → **4: Statistics** → **1: Stat Calculations** → **1: One-Variable Statistics**, Num of lists 1, X1 List: **a[]**, Frequency List: **b[]**. Read $\bar{x}=\mathrm{E}(X)$. $\Rightarrow \bar{x}=1.85$.

2 ← question *Worksheet · Q2*

(a) $0.2+0.1+r=0.6$ **M1** $\Rightarrow \boxed{r=0.3}$ **A1**

(b) Sum: $p+q+0.1+0.2+0.1+0.3=1 \Rightarrow p+q=0.3$ **M1**

$\mathrm{E}(X)$: $0p+1q+2(0.1)+3(0.2)+4(0.1)+5(0.3)=2.9$ **M1**

$q+2.7=2.9 \Rightarrow \boxed{q=0.2}$, $\boxed{p=0.1}$ **A1 A1**

3 answer → Calculator **Worksheet · Q5**

[Maximum mark: 6]

A random variable X has a probability distribution given in the following table.

x	-2	-1	0	1	2
$\mathrm{P}(X=x)$	p	$5p^2$	0.35	0.15	0.1

(a) Find the value of p. [3]

(b) Find $\mathrm{E}(X)$. [3]

4 answer → No calculator **Practice**

[Maximum mark: 5]

The probability distribution of a discrete random variable X is given below.

x	1	2	3	4	5
$\mathrm{P}(X=x)$	0.15	$2k$	0.25	k	0.15

(a) Find the value of k. [2]

(b) Find $\mathrm{E}(X)$. [3]

3 ← question *Worksheet · Q5*

(a) $p+5p^2+0.35+0.15+0.1=1 \Rightarrow 5p^2+p-0.4=0$ **M1**

$p=0.2$ or $p=-0.4$ **A1**

$p \geq 0$, so $\boxed{p=0.2}$ (reject -0.4: a probability cannot be negative) **R1**

(b) $5p^2=0.2$. $\mathrm{E}(X)=-2(0.2)-1(0.2)+0+1(0.15)+2(0.1)$ **M1 A1**

$=\boxed{-0.25}$ **A1**

TI-Nspire CX: Question 3

Solving for the unknown: home → **Add Calculator**, menu → **3: Algebra** → **1: Solve**, e.g. solve$(5p^2+p-0.4=0,\, p) \Rightarrow p=-0.4$ or 0.2. Then use Lists & Spreadsheet for E(X) (column A: $-2,-1,0,1,2$; column B: $0.2, 0.2, 0.35, 0.15, 0.1$).

4 ← question *Practice*

(a) $0.55+3k=1$ **M1** $\Rightarrow \boxed{k=0.15}$ **A1**

(b) $\mathrm{E}(X)=1(0.15)+2(0.3)+3(0.25)+4(0.15)+5(0.15)$ **M1 A1**

$=\boxed{2.85}$ **A1**

5 answer → No calculator **Practice**

[Maximum mark: 5]

The probability distribution of a discrete random variable X is given by $P(X = x) = kx$, for $x \in \{1, 2, 3, 4\}$.

(a) Find the value of k. [2]

(b) Find $E(X)$. [3]

6 answer → No calculator **Practice**

[Maximum mark: 6]

A discrete random variable X has probability distribution $P(X = x) = k(x^2 + 1)$, for $x = 0, 1, 2, 3$.

(a) Show that $k = \frac{1}{18}$. [2]

(b) Find $P(X \geq 2)$. [2]

(c) Find $E(X)$. [2]

5 ← question *Practice*

(a) $k(1+2+3+4)=1 \Rightarrow 10k=1$ **M1** $\boxed{k=\frac{1}{10}}$ **A1**

(b) $\mathrm{E}(X)=1\cdot\frac{1}{10}+2\cdot\frac{2}{10}+3\cdot\frac{3}{10}+4\cdot\frac{4}{10}$ **M1 A1**

$=\frac{30}{10}=\boxed{3}$ **A1**

6 ← question *Practice*

(a) $k(1+2+5+10)=1$ **M1** $\Rightarrow 18k=1,\ k=\frac{1}{18}$ ✓ **A1**

(b) $\frac{5}{18}+\frac{10}{18}=\boxed{\frac{15}{18}=\frac{5}{6}}$ **M1 A1**

(c) $\mathrm{E}(X)=\frac{0(1)+1(2)+2(5)+3(10)}{18}=\frac{42}{18}=\boxed{\frac{7}{3}}$ **M1 A1**

7 answer → No calculator **Practice**

[Maximum mark: 6]

The random variable X has the probability distribution shown.

x	1	2	3	4
$P(X = x)$	0.2	p	q	0.3

Given that $E(X) = 2.7$, find the value of p and the value of q. [6]

8 answer → No calculator **Practice**

[Maximum mark: 4]

The random variable X has the distribution shown, where a is a constant.

x	a	2	5
$P(X = x)$	0.3	0.5	0.2

Given that $E(X) = 3.1$, find the value of a. [4]

7 ← question *Practice*

Sum: $0.2+p+q+0.3=1 \Rightarrow p+q=0.5$ **M1 A1**

E(X): $0.2+2p+3q+1.2=2.7 \Rightarrow 2p+3q=1.3$ **M1 A1**

Solve: $2p+3(0.5-p)=1.3 \Rightarrow p=0.2$ **M1**

$\boxed{p=0.2,\ q=0.3}$ **A1**

8 ← question *Practice*

$0.3a+2(0.5)+5(0.2)=3.1$ **M1 A1**

$0.3a+2=3.1 \Rightarrow 0.3a=1.1$ **A1**

$\boxed{a=\frac{11}{3}}$ **A1**

9 answer →

Calculator **Practice**

[Maximum mark: 6]

A random variable X has the probability distribution shown.

x	0	1	2	3
$\mathrm{P}(X = x)$	0.1	$2p^2$	0.3	p

(a) Find the value of p. [3]

(b) Find $\mathrm{E}(X)$. [3]

10 answer →

Calculator **Practice**

[Maximum mark: 6]

The number of goals, X, scored by a team in a match has the distribution below.

x	0	1	2	3	4
$\mathrm{P}(X = x)$	0.12	0.27	0.30	k	0.08

(a) Find k. [1]

(b) Find $\mathrm{P}(X \geq 2)$. [2]

(c) Find the expected number of goals per match, and the expected total number of goals in a season of 38 matches. [3]

9 ← question *Practice*

(a) $0.1+2p^2+0.3+p=1 \Rightarrow 2p^2+p-0.6=0$ **M1**

$p=0.352$ or $p=-0.852$ **A1**

$p>0$: $\boxed{p=0.352}$ (3 s.f.) **A1**

(b) $2p^2=0.248$. $\mathrm{E}(X)=0.248+2(0.3)+3(0.352)$ **M1 A1**

$=\boxed{1.90}$ (3 s.f.) **A1**

TI-Nspire CX: Question 9

Solving for the unknown: home → **Add Calculator**, menu → **3: Algebra** → **1: Solve**, e.g. solve$(2p^2+p-0.6=0, p) \Rightarrow p=0.35208$. Store it: ctrl var (sto→) as p, then E(X) = $2p^2+0.6+3p$ enter.

10 ← question *Practice*

(a) $\boxed{k=0.23}$ **A1**

(b) $0.30+0.23+0.08=\boxed{0.61}$ **M1 A1**

(c) $\mathrm{E}(X)=0.27+0.60+0.69+0.32=\boxed{1.88}$ goals **M1 A1**

Season: $38\times 1.88=\boxed{71.4}$ goals **A1**

TI-Nspire CX: Question 10

Expected value: home → **Add Lists & Spreadsheet**. Type the x-values in column A and the probabilities in column B. Then menu → **4: Statistics** → **1: Stat Calculations** → **1: One-Variable Statistics**, Num of lists 1, X1 List: **a[]**, Frequency List: **b[]**. Read $\bar{x}$ = E(X). $\Rightarrow \bar{x}=1.88$. Then 38×1.88 on a Calculator page.

11 answer →

Calculator RV · Q12

[Maximum mark: 5]

A random variable X has a probability distribution given in the following table.

x	1	2	3	4	5
$P(X=x)$	$5a$	$4b$	0.3	0.15	0.05

Given the expected value of X is 2.65, find the value of a and of b.

12 answer →

No calculator Oxford Exam-style · Q17

[Maximum mark: 5]

The table shows the probability distribution of a discrete random variable X.

x	0	1	2	3
$P(X=x)$	0.2	$k+0.25$	$k-0.05$	0.3

(a) Find the value of k. [3]

(b) Hence calculate E(X). [2]

11 ← question *RV · Q12*

$\sum P = 1$: $5a+4b+0.5=1 \Rightarrow 5a+4b=0.5$ **M1 A1**

E(X): $5a+8b+0.9+0.6+0.25=2.65 \Rightarrow 5a+8b=0.9$ **M1 A1**

$4b=0.4$: $\boxed{b=0.1}$, $\boxed{a=0.02}$ **A1**

TI-Nspire CX: Question 11

Two equations: menu → **3: Algebra** → **2: Solve System of Linear Equations**, e.g. $m+z_1 s=x_1$, $m+z_2 s=x_2$.

12 ← question *Oxford Exam-style · Q17*

(a) $0.2+k+0.25+k-0.05+0.3=1 \Rightarrow 2k=0.3 \Rightarrow \boxed{k=0.15}$ **M1 A1**

A1

(b) $0(0.2)+1(0.4)+2(0.1)+3(0.3)=\boxed{1.5}$ **M1 A1**

13 answer →

Calculator **Oxford Exam-style · Q18**

[Maximum mark: 7]

An emergency call centre has four service lines that operate 24 hours every day of the week. The random variable X represents the number of phone lines in use during any five-minute period. Based on data collected over a long period of time, the centre manager has determined the following:

x	0	1	2	3	4
$\mathrm{P}(X = x)$	0.05	0.22	0.27	a	b

(a) State a condition on a and b. [2]

(b) Given that $\mathrm{E}(X) = 2.46$, determine the value of a and the value of b. [5]

14 answer →

No calculator **Oxford Ex 14B · Q3**

[Maximum mark: 3]

The discrete random variable X has probability distribution $\mathrm{P}(x) = \frac{x}{36}$ for $x = 1, 2, 3, \ldots, 8$. Find $\mathrm{E}(X)$.

13 ← question *Oxford Exam-style · Q18*

(a) $\boxed{a+b=0.46}$ (and $a,b \geq 0$) **M1 A1**

(b) $0.22+0.54+3a+4b=2.46 \Rightarrow 3a+4b=1.7$ **M1 A1**

$\boxed{a=0.14}$, $\boxed{b=0.32}$ **M1 A1 A1**

TI-Nspire CX: Question 13

Two equations: menu → **3: Algebra** → **2: Solve System of Linear Equations**, e.g. $m+z_1 s=x_1$, $m+z_2 s=x_2$.

14 ← question *Oxford Ex 14B · Q3*

$$\mathrm{E}(X)=\sum x\cdot\frac{x}{36}=\frac{1+4+9+\cdots+64}{36}=\frac{204}{36}$$ **M1 A1**

$$=\boxed{\frac{17}{3}=5\frac{2}{3}}$$ **A1**

15 answer → No calculator **Oxford Ex 14B · Q4**

[Maximum mark: 5]

For the discrete random variable X, the probability distribution is given by

$$\mathrm{P}(X=x)=\begin{cases} kx & x=1,2,3,4,5 \\ k(10-x) & x=6,7,8,9 \end{cases}$$

Find:

(a) the value of the constant k [3]

(b) $\mathrm{E}(X)$. [2]

15 ← question *Oxford Ex 14B · Q4*

(a) $k(1+2+3+4+5)+k(4+3+2+1)=25k=1 \Rightarrow \boxed{k=\frac{1}{25}}$ **M1**

A1 A1

(b) $\frac{1}{25}(1+4+9+16+25+24+21+16+9)=\frac{125}{25}=\boxed{5}$ (also by symmetry about 5) **M1 A1**

16 answer →

No calculator

Oxford Ex 14B · Q5

[Maximum mark: 5]

Complete this probability distribution, in terms of k, for a discrete random variable X.

x	1	2	3
$\mathrm{P}(X = x)$	0.2	$1-k$	...

(a) What range of values can k take? Give your answer in the form $a \leq k \leq b$, $a, b \in \mathbb{Q}$. [3]

(b) Find the mean of the distribution in terms of k. [2]

16 ← question *Oxford Ex 14B · Q5*

$\mathrm{P}(X=3) = 1-0.2-(1-k) = \boxed{k-0.2}$ **A1**

(a) $k-0.2 \geq 0$ and $1-k \geq 0$: $\boxed{0.2 \leq k \leq 1}$ **M1 A1**

(b) $0.2+2(1-k)+3(k-0.2) = \boxed{1.6+k}$ **M1 A1**

17 answer → Calculator **RV · Q51**

[Maximum mark: 15]

Tom enters a tennis competition where he plays three matches every day. The results of each match are independent of each other. Let A be the number of matches Tom wins on any given day of the competition.

a	0	1	2	3
P$(A = a)$	0.15	0.2	p	0.25

(a) Find the value of p. [2]

(b) (i) A day is chosen at random. Write down the probability that Tom wins every match he plays. (ii) The competition goes for four days. Find the probability that Tom wins every match on exactly three of these days. [4]

Clare enters the same tennis competition. Let B be the number of matches Clare wins on any given day.

b	0	1	2	3
P$(B = b)$	0.05	0.1	0.35	0.5

(c) Find E$[B]$. [3]

On the final day, both Tom and Clare play their three matches and their results are independent. The number of wins Tom and Clare obtain are added together to form a total of six.

(d) (i) Find the probability that they win more than four matches combined. (ii) Given that they win more than four matches combined, find the probability that Clare won all of her matches. [6]

17 ← question *RV · Q51*

(a) $p = 1 - 0.6 = \boxed{0.4}$ **M1 A1**

(b)(i) $\boxed{0.25}$ (ii) $W \sim \text{B}(4, 0.25)$: $\text{P}(W = 3) = 4(0.25)^3(0.75) = \boxed{0.0469}$

A1 M1 M1 A1

(c) $0.1 + 0.7 + 1.5 = \boxed{2.3}$ **M1 A2**

(d)(i) total 5 or 6: $(2,3), (3,2), (3,3)$: $0.4(0.5) + 0.25(0.35) + 0.25(0.5) = \boxed{0.4125}$ **M1 M1 A1**

(ii) $\dfrac{0.2 + 0.125}{0.4125} = \boxed{0.788}$ **M1 M1 A1**

TI-Nspire CX: Question 17

menu → **5: Probability** → **5: Distributions** → **D: Binomial Pdf**: n, p, X value (one value).

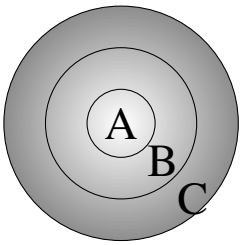
```
binomPdf(4,0.25,3)
```

18 answer →

No calculator **Worksheet · Q3**

[Maximum mark: 7]

The following diagram shows an archery target which is divided into three regions A, B and C.



A contest consists of an archer shooting one arrow at the target. The probability of hitting each region is given in the following table.

Region	A	B	C
Probability	$\frac{1}{24}$	$\frac{4}{24}$	$\frac{7}{24}$

(a) Find the probability that the arrow does **not** hit the target. [3]

An archer scores points on the contest as shown in the following table.

Region	A	B	C	Outside
Points	10	6	k	-4

(b) Given that the contest is fair, find the value of k. [4]

18 ← question *Worksheet · Q3*

(a) $\mathrm{P(hit)} = \dfrac{1+4+7}{24} = \dfrac{12}{24}$ **M1 A1**

$\mathrm{P(miss)} = 1 - \dfrac{1}{2} = \boxed{\dfrac{1}{2}}$ **A1**

(b) Fair game $\Rightarrow$ E(points) $= 0$ **R1**

$10 \cdot \dfrac{1}{24} + 6 \cdot \dfrac{4}{24} + k \cdot \dfrac{7}{24} - 4 \cdot \dfrac{1}{2} = 0$ **M1**

$\dfrac{34+7k}{24} = 2 \Rightarrow 34 + 7k = 48$ **A1**

$\boxed{k = 2}$ **A1**

19 answer →

Calculator **Worksheet · Q4**

[Maximum mark: 5]

A biased coin has a probability of showing "heads" of 0.7. Jim pays \$20 to play a game where he flips the coin. He gets no money if he flips a "heads" and gets \$$k$ if he flips a "tails". The game is fair.

Find the value of k. [5]

20 answer →

No calculator **Practice**

[Maximum mark: 5]

A player pays \$$k$ to roll a fair six-sided die. If the die shows a 6 the player receives \$12; if it shows a 4 or a 5 the player receives \$3; otherwise the player receives nothing.

(a) Find the value of k for which the game is fair. [3]

(b) The player is charged \$4 to play. Find the expected profit or loss for the player per game. [2]

19 ← question *Worksheet · Q4*

Let W be the amount Jim receives. $P(T) = 1 - 0.7 = 0.3$ **A1**

outcome	H	T
W	0	k
P	0.7	0.3

Fair: E(amount received) = cost to play **R1**

$0(0.7) + k(0.3) = 20$ **M1 A1**

$k = \dfrac{20}{0.3} = \boxed{66.7}$ ($66.67) **A1**

TI-Nspire CX: Question 19

home → **Add Calculator**: $20 \div 0.3$ enter. For a decimal answer press ctrl enter ⇒ 66.6667.

20 ← question *Practice*

(a) E(received) $= 12 \cdot \dfrac{1}{6} + 3 \cdot \dfrac{2}{6} + 0 \cdot \dfrac{3}{6}$ **M1** $= 2 + 1 = 3$ **A1**

Fair ⇒ $\boxed{k = 3}$ **A1**

(b) E(profit) $= 3 - 4 = \boxed{-\$1}$, i.e. an expected loss of \$1 per game **M1 A1**

21 answer → No calculator **Practice**

[Maximum mark: 5]

Three fair coins are tossed. A player wins \$10 if all three show heads, wins \$2 if exactly two show heads, and loses \$4 otherwise.

(a) Write down the probability distribution of the player's gain G. [3]

(b) Show that the game is fair. [2]

22 answer → No calculator **Practice**

[Maximum mark: 6]

A dart is thrown at a board with three regions A, B and C. The probabilities of hitting A, B and C are k, $2k$ and $3k$ respectively, and the probability of missing the board is 0.4.

(a) Find k. [2]

Points are scored as follows: A: 10, B: 5, C: 2, miss: -1.

(b) Find the expected score for one throw. [3]

(c) Find the expected total score for 20 throws. [1]

21 ← question *Practice*

(a)

g	10	2	-4
$\mathrm{P}(G=g)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{4}{8}$

A1 A1 A1

(b) $\mathrm{E}(G)=\frac{10}{8}+\frac{6}{8}-\frac{16}{8}$ **M1** $=0$, so the game is fair **A1**

22 ← question *Practice*

(a) $6k+0.4=1$ **M1** $\boxed{k=0.1}$ **A1**

(b) $\mathrm{E}=10(0.1)+5(0.2)+2(0.3)-1(0.4)$ **M1 A1** $=\boxed{2.2}$ **A1**

(c) $20\times 2.2=\boxed{44}$ **A1**

23 answer →

Calculator **Practice**

[Maximum mark: 5]

An insurance company sells a one-year policy for \$350. The probability that a policy holder makes a claim in the year is 0.012, and each claim costs the company \$25 000. (Assume at most one claim per policy.)

(a) Find the company's expected profit per policy. [3]

(b) The company sells 2000 policies. Find its expected total profit. [2]

24 answer →

No calculator **Oxford Ex 14B · Q8**

[Maximum mark: 5]

Suppose an instant lottery ticket is purchased for \$2. The possible prizes are \$0, \$2, \$20, \$200 and \$1000. Let Z be the random variable representing the amount won on the ticket, and suppose Z has the following distribution.

z	0	2	20	200	1000
$\mathrm{P}(Z = z)$		0.2	0.05	0.001	0.0001

(a) Determine $\mathrm{P}(Z = 0)$. [1]

(b) Determine $\mathrm{E}(Z)$ and interpret its meaning. [2]

(c) How much should you expect to gain or lose on average per ticket? [2]

23 ← question *Practice*

(a) Profit: \$350 with prob. 0.988; $350 - 25000 = -\$24\,650$ with prob. 0.012 **M1**

$E = 350(0.988) - 24650(0.012)$ **M1** $= \boxed{\$50}$ **A1**

(or $350 - 0.012 \times 25000 = 50$)

(b) $2000 \times 50 = \boxed{\$100\,000}$ **M1 A1**

TI-Nspire CX: Question 23

home → **Add Calculator**: $350 \times 0.988 - 24650 \times 0.012$ enter ⇒ 50.

24 ← question *Oxford Ex 14B · Q8*

(a) $1 - 0.2511 = \boxed{0.7489}$ **A1**

(b) $0.4 + 1 + 0.2 + 0.1 = \boxed{1.7}$: on average \$1.70 is won per ticket **M1 A1**

(c) $1.70 - 2 = -0.30$: **lose \$0.30** per ticket **M1 A1**

25 answer → No calculator **RV · Q40**

[Maximum mark: 11]

Carmen plays a game at a stall at a local fair. The stall owner lays n cards face down on a table. Three of the cards say "Win" on the hidden side, the others say "Try Again". Carmen draws cards from the table, one after the other, without replacement. The game ends when Carmen draws a card that says "Win".

(a) Find the probability, in terms of n, that the game will end on her 2nd draw. [2]

(b) Given that $n = 8$, find the probability that the game will end on her 6th draw. [2]

Carmen plays the game when $n = 8$. She pays \$6 to play and can earn money back depending on the number of draws it takes to obtain a card that says "Win". She earns no money back if she obtains a card that says "Win" on her 6th draw. Let M be the amount of money that she earns back playing the game.

Number of draws	1	2	3	4	5	6
Money earned back (\$$M$)	10	5	4	$\frac{3k}{4}$	$\frac{k}{3}$	0

(c) Find the value of k so that this is a fair game. [7]

25 ← question *RV · Q40*

(a) $\dfrac{n-3}{n} \times \dfrac{3}{n-1} = \boxed{\dfrac{3(n-3)}{n(n-1)}}$ **M1 A1**

(b) five "Try Again" first: $\dfrac{5}{8} \cdot \dfrac{4}{7} \cdot \dfrac{3}{6} \cdot \dfrac{2}{5} \cdot \dfrac{1}{4} \cdot 1 = \boxed{\dfrac{1}{56}}$ **M1 A1**

(c) P: $\dfrac{3}{8}, \dfrac{15}{56}, \dfrac{5}{28}, \dfrac{3}{28}, \dfrac{3}{56}, \dfrac{1}{56}$ **M1 A2**

fair: $\mathrm{E}(M) = 6$: $\dfrac{30}{8} + \dfrac{75}{56} + \dfrac{20}{28} + \dfrac{3k}{4} \cdot \dfrac{3}{28} + \dfrac{k}{3} \cdot \dfrac{3}{56} = 6$ **M1 A1**

$\dfrac{325}{56} + \dfrac{11k}{112} = 6 \Rightarrow \boxed{k = 2}$ **A1 A1**

26 answer → No calculator **School HW Extension · Q5**

[Maximum mark: 15]

A bag contains n marbles, two of which are green. Claudia plays a game in which she randomly draws marbles out of the bag, one after another, without replacement. The game ends when Claudia draws a green marble.

(a) Find the probability, in terms of n, that the game will end on her: (i) 1st draw; (ii) 2nd draw. [4]

(b) Given that $n = 6$, find the probability that the game will end on her: (i) 3rd draw; (ii) 5th draw. [4]

Claudia plays the game when $n = 6$. She pays \$11 to play and can earn money back depending on the number of draws it takes to obtain a green marble. She earns no money back if she obtains a green marble on her 5th draw. Let M be the amount of money that she earns back playing the game.

Number of draws	1	2	3	4	5
Money earned back (\$$M$)	15	$4k$	10	$2k$	0

(c) Find the value of k so that this is a fair game. [7]

26 ← question *School HW Extension · Q5*

(a)(i) $\boxed{\frac{2}{n}}$ (ii) $\frac{n-2}{n}\cdot\frac{2}{n-1}=\boxed{\frac{2(n-2)}{n(n-1)}}$ **A1 M1 A2**

(b)(i) $\frac{4}{6}\cdot\frac{3}{5}\cdot\frac{2}{4}=\boxed{\frac{1}{5}}$ (ii) $\frac{4}{6}\cdot\frac{3}{5}\cdot\frac{2}{4}\cdot\frac{1}{3}\cdot 1=\boxed{\frac{1}{15}}$ **M1 A1 M1 A1**

(c) P: $\frac{1}{3},\ \frac{4}{15},\ \frac{1}{5},\ \frac{2}{15},\ \frac{1}{15}$ **M1 A2**

$15\cdot\frac{1}{3}+4k\cdot\frac{4}{15}+10\cdot\frac{1}{5}+2k\cdot\frac{2}{15}=11\Rightarrow 7+\frac{4k}{3}=11$ **M1 A1**

$\boxed{k=3}$ **A1 A1**

27 answer → No calculator **Worksheet · Q6**

[Maximum mark: 6]

Two unbiased dice are thrown and the scores are recorded. Let the random variable X be the minimum of these two scores. The probability distribution of X is given in the following table.

x	1	2	3	4	5	6
$\mathrm{P}(X = x)$	$\frac{11}{36}$	p	$\frac{7}{36}$	q	$\frac{3}{36}$	$\frac{1}{36}$

(a) Find the value of p and the value of q. [3]

(b) Find the expected value of X. [3]

27 ← question *Worksheet · Q6*

(a) Sample space (minimum of the two scores):

min	1	2	3	4	5	6
1	1	1	1	1	1	1
2	1	**2**	**2**	**2**	**2**	**2**
3	1	**2**	3	3	3	3
4	1	**2**	3	4	4	4
5	1	**2**	3	4	5	5
6	1	**2**	3	4	5	6

M1

Count the 2s: $\boxed{p = \frac{9}{36}}$; count the 4s: $\boxed{q = \frac{5}{36}}$ **A1 A1**

(check: $11+9+7+5+3+1 = 36$)

(b) $\mathrm{E}(X) = \dfrac{1(11)+2(9)+3(7)+4(5)+5(3)+6(1)}{36}$ **M1 A1** $= \boxed{\frac{91}{36}} \approx 2.53$ **A1**

28 answer →

 No calculator

Worksheet · Q7

[Maximum mark: 6]

The faces of a fair six-sided die are numbered 2, 4, 4, 4, 6, 6. Let X be the discrete random variable that models the sum of scores obtained when two such dice are rolled.

(a) Complete the probability distribution table for X. [3]

x	4	6	8	10	12
$\mathrm{P}(X = x)$					

(b) Find the expected value of X. [3]

28 ← question *Worksheet · Q7*

(a) Sample space of sums:

+	2	4	4	4	6	6
2	4	6	6	6	8	8
4	6	8	8	8	10	10
4	6	8	8	8	10	10
4	6	8	8	8	10	10
6	8	10	10	10	12	12
6	8	10	10	10	12	12

M1

x	4	6	8	10	12
P	$\frac{1}{36}$	$\frac{6}{36}$	$\frac{13}{36}$	$\frac{12}{36}$	$\frac{4}{36}$

A2

(b) $\mathrm{E}(X) = \dfrac{4+36+104+120+48}{36}$ **M1 A1** $= \dfrac{312}{36} = \boxed{\dfrac{26}{3}} \approx 8.67$ **A1**

29 answer → No calculator **Practice**

[Maximum mark: 6]

Two fair four-sided dice, each numbered 1, 2, 3, 4, are rolled. Let X be the positive difference between the two scores.

(a) Show the possible outcomes in a sample-space table, and hence write down the probability distribution of X. [4]

(b) Find $\mathrm{E}(X)$. [2]

30 answer → No calculator **Practice**

[Maximum mark: 6]

A bag contains 5 red and 3 blue counters. Two counters are taken at random **without replacement**. Let X be the number of red counters taken.

(a) Find the probability distribution of X. [4]

(b) Find $\mathrm{E}(X)$. [2]

29 ← question *Practice*

(a)

	1	2	3	4
1	0	1	2	3
2	1	0	1	2
3	2	1	0	1
4	3	2	1	0

M1

x	0	1	2	3
P	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{2}{16}$

A3

(b) $\mathrm{E}(X) = \dfrac{0+6+8+6}{16} = \boxed{\dfrac{20}{16} = 1.25}$ **M1 A1**

30 ← question *Practice*

(a) $\mathrm{P}(X=0) = \dfrac{3}{8}\cdot\dfrac{2}{7} = \dfrac{6}{56},\ \mathrm{P}(X=1) = 2\cdot\dfrac{5}{8}\cdot\dfrac{3}{7} = \dfrac{30}{56},\ \mathrm{P}(X=2) = \dfrac{5}{8}\cdot\dfrac{4}{7} = \dfrac{20}{56}$ **M1 A1 A1 A1**

(check: $6+30+20=56$)

(b) $\mathrm{E}(X) = \dfrac{0+30+40}{56} = \dfrac{70}{56} = \boxed{1.25}$ **M1 A1**

31 answer →

No calculator **Oxford Exam-style · Q16**

[Maximum mark: 5]

There are ten black and eight white socks in a drawer. Jeannie, without looking, takes out a sock at random, looks at its colour and puts it back in the drawer. Then, again without looking, takes another sock out. Let X be the random variable that represents the number of black socks taken out by Jeannie.

(a) State the possible values of the random variable X. [1]

(b) Find $\mathrm{P}(X = 2)$. [2]

(c) Construct the probability distribution table for the random variable X. [2]

32 answer →

No calculator **Oxford Ex 14B · Q6**

[Maximum mark: 7]

Ten balls of identical size are in a bag. Two of the balls are red and the rest are blue. Balls are picked out at random from the bag and are not replaced. Let R be the number of balls drawn out, up to and including the first red one.

(a) List the possible values of R and their associated probabilities. [4]

(b) Calculate the mean value of R. [2]

(c) What is the most likely value of R? [1]

31 ← question *Oxford Exam-style · Q16*

(a) $\boxed{0, 1, 2}$ **A1**

(b) $\dfrac{10}{18} \times \dfrac{10}{18} = \boxed{\dfrac{25}{81}}$ **M1 A1**

(c)

x	0	1	2
$\mathrm{P}(X = x)$	$\frac{16}{81}$	$\frac{40}{81}$	$\frac{25}{81}$

A2

32 ← question *Oxford Ex 14B · Q6*

(a) $R = 1, 2, \ldots, 9$ with $\mathrm{P}(R = r) = \dfrac{10 - r}{45}$: $\dfrac{9}{45}, \dfrac{8}{45}, \dfrac{7}{45}, \ldots, \dfrac{1}{45}$ **M1 A3**

(e.g. $\mathrm{P}(R = 2) = \frac{8}{10} \cdot \frac{2}{9} = \frac{8}{45}$)

(b) $\mathrm{E}(R) = \dfrac{\sum r(10 - r)}{45} = \dfrac{165}{45} = \boxed{3\dfrac{2}{3}}$ **M1 A1**

(c) $\boxed{R = 1}$ **A1**